Project Work: Advanced Drone Technology

Project Title: Recreating the Cessna 210 (NASA Variant) Using OpenVSP

Program: Winter Internship Training program – Advanced Drone Technology

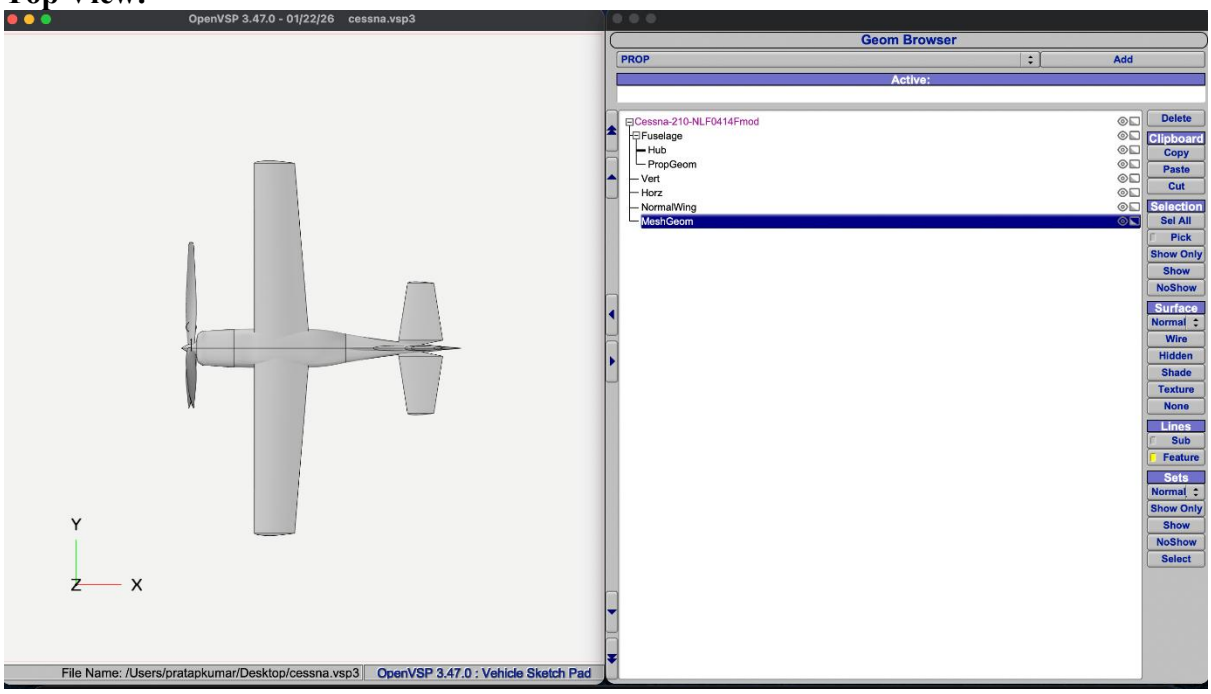
Institution: INDIA SPACE LAB

1. Overview

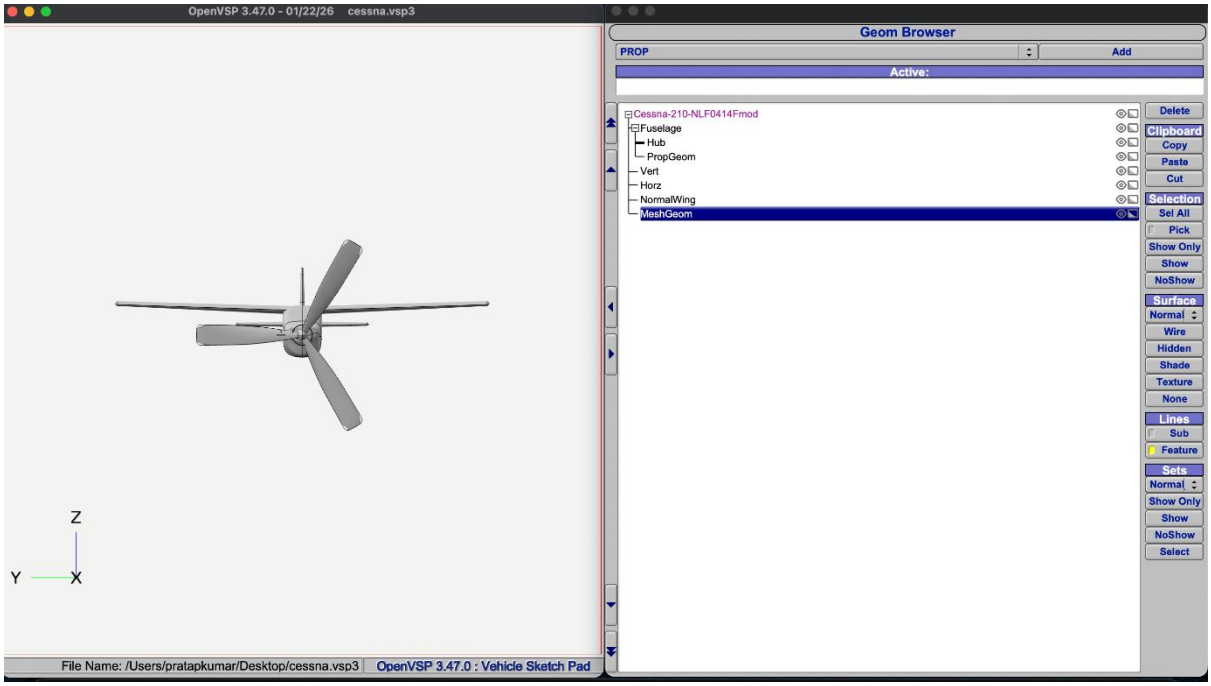
This project involves modelling a complete fixed-wing aircraft, specifically the Cessna 210, utilizing OpenVSP. The modelling process relied on geometric parameterization and analysing 3-view 2D drawings to construct a 3D digital representation.

2. 3-View Screenshots

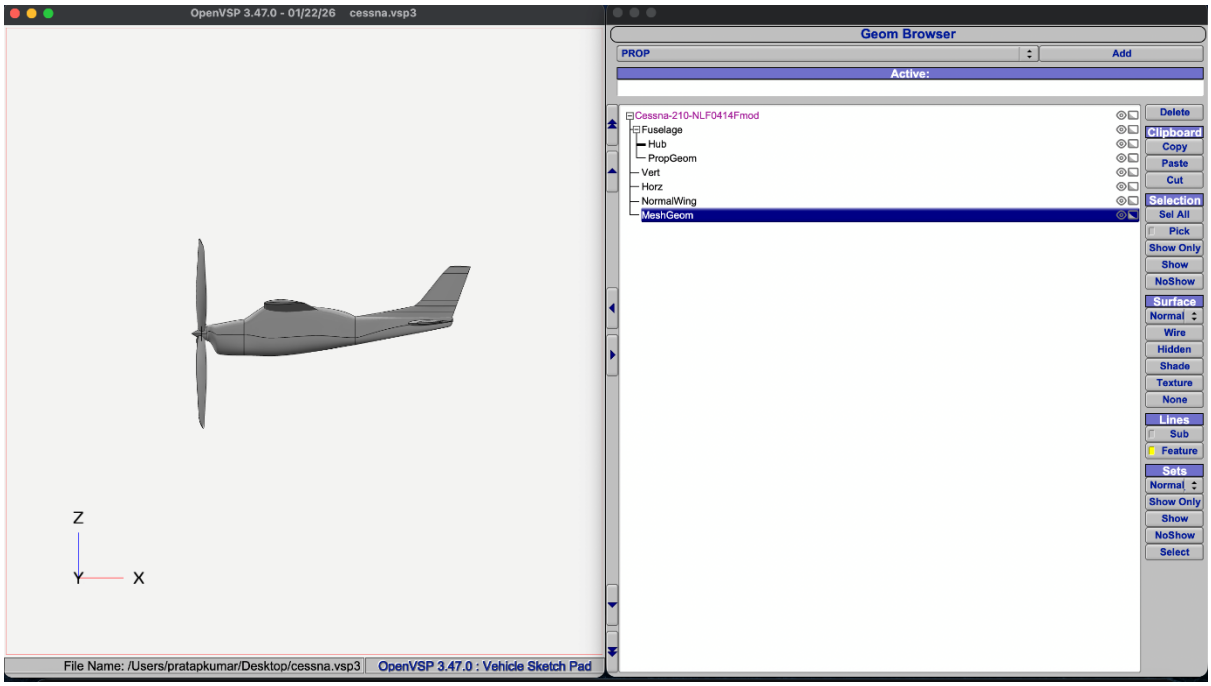
- **Top View:**



- **Front View:**



Side View:



3. Geometric Parameters & Sizing Methods

Wing Parameters Used

The main wing was constructed using the default OpenVSP Wing geometry.

- **Span:** The primary wing span was set to exactly 36.748 ft, as dimensioned in the front and top blueprints.
- **Air foil:** The default NACA air foil was utilized, which provides a suitable aerodynamic profile for this phase of modelling.

- **Positioning & Shaping:** The wing was translated along the Z-axis to replicate the Cessna 210's signature high-wing configuration. Parameters such as the root chord, tip chord, taper ratio, sweep angle, and dihedral angle were adjusted visually to ensure geometric proportion consistency with the provided 3-view reference images.

Fuselage Length

- **Overall Dimension:** The fuselage was modelled with an exact overall length of 24.580 ft, derived directly from the side-view blueprint.
- **Cross-Section Adjustments:** Using the XSec Editor, the tubular geometry was heavily modified across multiple cross-sections. The central indices were expanded to accommodate a maximum cabin width and a cabin height of approximately 3.814 ft spanning a length of 5.176 ft. The Z-offset was utilized to flatten the belly of the aircraft, while the forward and aft sections were adjusted to match the distinct nose curvature and tail cone taper of the Cessna.

Tail Sizing Method

The empennage (tail section) was constructed using two separate wing geometries scaled down and positioned at the rear of the fuselage.

- **Horizontal Tail:** Modelled with a span of 12.990 ft and a root chord of 2.616 ft. The X, Y, and Z placements were adjusted to seat it properly on the tapering fuselage.
- **Vertical Tail:** Created by rotating a wing geometry 90 degrees along the X-axis. The sizing was matched to the side view, showing an upper tail height of 2.94 ft, and aligned with the top-down projection showing a length of 4.379 ft. Sweep and root chord were adjusted to match the characteristic swept-back design.